

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – SIH26097

Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY

Theme – Agriculture, FoodTech & Rural Development

PS Category – Software

Team ID – _____

Team Name (Registered on portal) – _____



**SMART INDIA
HACKATHON
2026**

HunarMarg

हुनरमार्ग • the path from skill to livelihood

The gap

A form picks the course, not the person, so SC job seekers land in training that doesn't fit them.

41%

of certified trainees were placed
CAG 2025

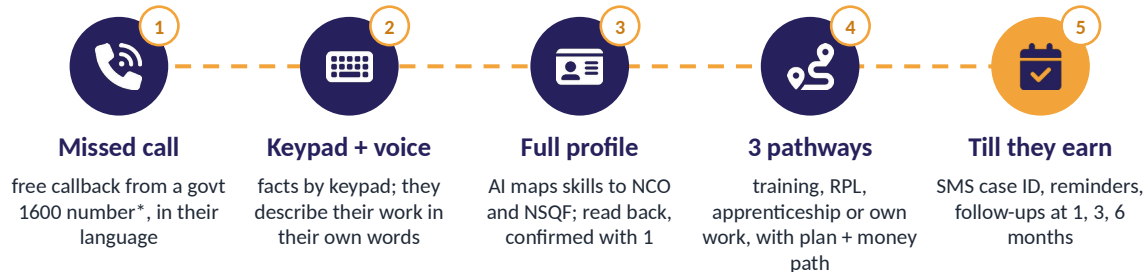
51.6%

of rural women have no phone of their own
NSO 2025

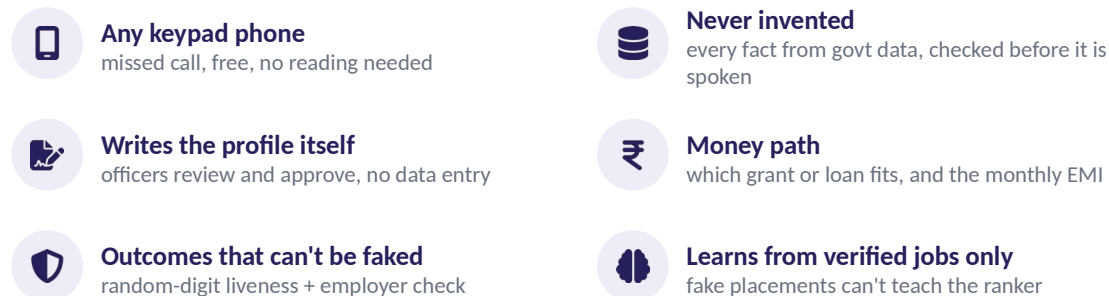
66.1%

SC literacy: forms don't reach many
Census 2011

The journey: one call, guided end to end



What only HunarMarg does



How HunarMarg works

During the call
seconds



After the call
days



AI understands → rules and data decide → facts checked → people approve

Stack

Calls Exotel • 1600 number* • SMS

AI (fine-tuned) IndicConformer • Indic Parler-TTS • Sarvam-30B • MuRIL • e5 • LightGBM • Satyavaani

Govt data NCO-2015 • NQR • schemes • DigiLocker* • AJAY app*

Backend



on Indian GPU cloud

Models we fine-tune

Model	What it does	How we train it	Must pass
IndicConformer-600M	speech → text, 22 languages	100 h consented phone calls per language + IndicVoices • ~72 GPU-h	Hindi error ≤ 20%
Indic Parler-TTS	voice, 21 languages	12 h of one local voice artist per language • ~24 GPU-h	listeners ≥ 4/5
Sarvam-30B	reads the story, fills a fixed form	LoRA on 5,000 story → form pairs checked by people • ~16 GPU-h	right job in top 3 ≥ 90%
multilingual-e5 + BM25	keyword + meaning job search	every “1” at read-back = right pair, “none” = wrong pair	right job in top 5 ≥ 90%
LightGBM LambdaRank	filters, then scores 6 factors	officer-set weights (AHP) first, then verified jobs at 6 months	gender gap ≤ 5 pts
Satyavaani (team's own)	liveness on follow-up calls	Indian phone speech + 3 voice-cloning tools; no voiceprint	flags, never rejects

Can it be built and run? Yes.

Proof it can work

22 cr

feature-phone users: the channel exists
Business Standard 2026

3 cr+

reached by Kilkari, a govt voice-call line;
answers 50% → 76% by the 3rd try
BMJ Global Health

22 • 21

languages in open Indian speech and voice
models
AI4Bharat

19.3%

best Indian speech error on IndicVoices:
our bar before we switch
Sarvam 2026



our prototype takes real calls in 3
languages and gives top 3 options
GitHub

What it costs

₹163

per person, all-in, at 50,000
a year

0.24%

of ₹69,200 GIA support per
person

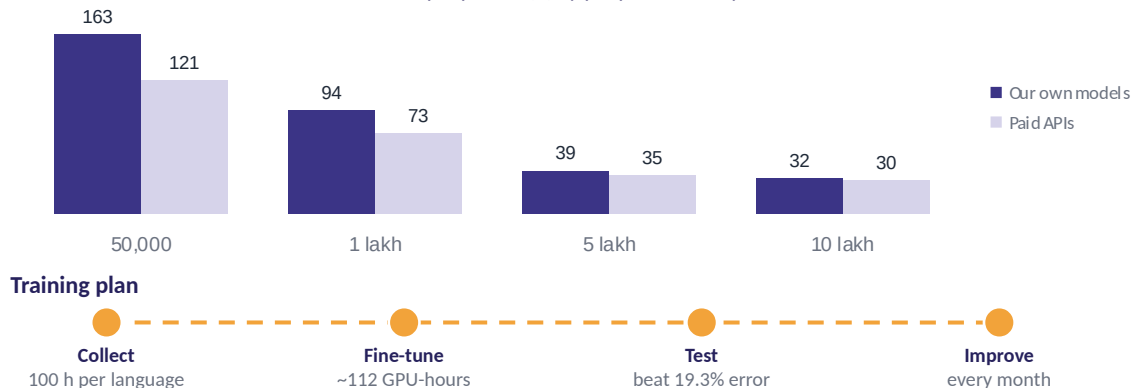
₹8.8 L

GPUs a year, in India

₹2 L

one time per language

Cost per person (₹) by people served a year



Risks and safeguards



Village dialects

keypad for facts, everything read back,
human fallback



AI says something wrong

only facts that match a govt data row are
spoken



Unknown numbers ignored

they call first; callback from a 1600
number*



Our model is weaker

switch a language only after it beats 19.3%;
API backup

What changes on the ground

41% → 70%

placed after training



CAG found 41%; 70% is the GIA target

62% → 73%

stay in the job



when told real pay and place first (J-PAL trial)

109 min

officer time saved per person

≈ 52 staff-years a year · our estimate

₹44,136

saved on one small-business loan

₹80,000 tailoring unit, right money path · our calculation

 ~50,000 GIA beneficiaries a year 47,000+ SC-majority villages already on the PM-AJAY dashboard

Who benefits, and how



SC job seeker

right course first time, free,
own language, no middleman



Women

keypad-only option, "safe to talk?", women-only batches



District officers

ready profiles, less paperwork,
one case record



Training centres

trainees who fit, fewer dropouts



Banks • NSFDC

better-matched loan applicants



Ministry

block-level demand for plans,
outcomes it can trust

Social fair, right-first-time advice · **Economic** 0.24% extra so the rest buys the right course · **Environment** no paper, fewer office trips

Where every number comes from

Government and research

- S1** Census 2011 via MoSJE Handbook 2021: SC literacy 66.1% socialjustice.gov.in
- S2** NSO Telecom survey 2025: phone ownership mospi.gov.in
- S3** MoSJE PM-AJAY factsheet, 28 Sep 2026: ₹1,730 cr, 2.5 lakh people pib.gov.in
- S4** PM-AJAY guidelines: 70% placement, 30% women socialjustice.gov.in
- S5** CAG Report No. 20 of 2025: 41% of 56 lakh placed cag.gov.in
- S6** NCO-2015 • NQR / NCVET • NSFDC • PM Vishwakarma nqr.gov.in
- S7** PIB: PM-AJAY portal and AJAY app, 26 May 2026 pib.gov.in
- S8** Business Standard, Apr 2026: 22 crore feature-phone users business-standard.com
- S9** BMJ Global Health: Kilkari reach and answer rates gh.bmj.com
- S10** Chakravorty et al., J-PAL: job information → retention 62% → 73% povertyactionlab.org

Technology, policy and our work

- S11** Sarvam: Saaras v3, 19.31% WER on IndicVoices sarvam.ai
- S12** AI4Bharat: IndicConformer, IndicVoices, Indic Parler-TTS huggingface.co/ai4bharat
- S13** Sarvam-30B open weights, Apache 2.0 huggingface.co/sarvamai
- S14** E2E Networks GPU prices: L4 ₹49/h, L40S ₹102/h e2enetworks.com
- S15** Sarvam API pricing; carrier rate cards (team research) docs.sarvam.ai
- S16** DPDP Rules 2025, notified 14 Nov 2025 meity.gov.in
- S17** TRAI: 1600 series for government-to-citizen calls traigov.in
- S18** Our work: prototype, MODELS.md, cost_model.py, measure.py [GitHub](https://github.com)


GitHub

code + docs/MODELS.md


Prototype

number + PIN (add)


Demo video

a real call (add link)


Calculations

scripts/cost_model.py